CHARUSAT University

Subject: Machine learning Total Marks: 30

Section - I: Multiple Choice Questions

1. What happens when a model is too complex and performs well on the training data but poorly on new unseen data?

A) Underfitting occurs

B) Overfitting occurs

C) Model is biased

D) Model is robust

2. Which of the following is a characteristic of a model that is underfitting the training data?

A) The model is too complex and fits the noise in the data.

B) The model is too simple and fails to capture the underlying patterns in the data.

C) The model has a high bias and a low variance.

D) The model has a low bias and a high variance.

3. What is a common characteristic of a model that is underfitting the training data?

A) The model performs well on the training data but poorly on new unseen data.

B) The model is too complex and performs well on the training data.

C) The model fails to capture the underlying patterns in the training data.

D) The model is simple and performs poorly on the training data.

4. What is the primary difference between overfitting and underfitting in machine learning models?

A) Overfitting occurs when the model is too complex, while underfitting occurs when the model is too simple.

B) Overfitting occurs when the model is too simple, while underfitting occurs when the model is too complex.

C) Overfitting occurs when the model has high bias, while underfitting occurs when the model has high variance.

D) Overfitting occurs when the model has high variance, while underfitting occurs when the model has high bias.

5. What is the primary difference between overfitting and underfitting in machine learning?

A) Overfitting occurs when the model is too complex, while underfitting occurs when the model is too simple.

B) Overfitting occurs when the model is too simple, while underfitting occurs when the model is too complex.

C) Overfitting occurs when the training data is too small, while underfitting occurs when the training data is too large.

D) Overfitting occurs when the model is trained for too long, while underfitting occurs when the model is trained for too short.

Section - II: Descriptive / Case Study-Based Questions

1. A marketing company, 'TargetPro', is developing a machine learning model to predict customer churn based on various features such as age, location, and purchase history. The model is trained on a dataset of 1000 customers, with 700 used for training and 300 for testing. The training data is split into two subsets: 80% for model training and 20% for validation. The team tries two different models: a simple linear regression model and a complex neural network model. The results are as follows: Linear Regression Model: - Training accuracy: 70% - Validation accuracy: 68% - Testing accuracy: 65% Neural Network Model: - Training accuracy: 95% - Validation accuracy: 82% - Testing accuracy: 75% Which model is more likely to be suffering from overfitting, and which one from underfitting? Explain your reasoning, considering the performance metrics of both models.

2. A marketing company, "SmartAd", uses machine learning models to predict the click-through rate (CTR) of online advertisements. They have a dataset of 10,000 advertisements, each with 10 features such as ad content, target audience, and display location. The company's data scientist, Rachel, is tasked with developing a model to predict CTR. Rachel trains three different machine learning models: a simple linear regression model, a decision tree model, and a complex neural network model. The performance metrics for each model on the training and testing datasets are as follows: Model 1 (Linear Regression): Training dataset - Mean Squared Error (MSE) = 0.05, R-squared = 0.8 Testing dataset - MSE = 0.15, R-squared = 0.5 Model 2 (Decision Tree): Training dataset - MSE = 0.01, R-squared = 0.95 Testing dataset - MSE = 0.12, R-squared = 0.7 Model 3 (Neural Network): Training dataset - MSE = 0.005, R-squared = 0.98 Testing dataset - MSE = 0.20, R-squared = 0.4 Which model is likely suffering from overfitting, and which model is likely suffering from underfitting? Explain your reasoning, and recommend a course of action for Rachel to improve the performance of the models.

3. A marketing company, Green Earth, wants to develop a machine learning model to predict the sales of their new eco-friendly product based on various features such as price, advertising expenditure, seasonality, and competitor pricing. They collect a dataset of 1000 instances, each representing a month of sales data. After splitting the data into training and testing sets, they train a model using a polynomial regression algorithm with a degree of 10. The model performs extremely well on the training data, with an R-squared value of 0.95. However, when they use the model to make predictions on the testing data, the R-squared value drops to 0.4. Explain the possible reason for this discrepancy in performance and suggest a potential solution to improve the model's predictive ability on unseen data.

4. A company that specializes in recommending personalized fitness routines to its clients has collected a large dataset of user information, including age, height, weight, and exercise habits. They want to develop a machine learning model that can predict the ideal fitness routine for a new user based on their characteristics. The company's data scientist, Rachel, has collected a dataset of 1000 users and has developed two different models to make these predictions. Model A, a simple linear regression model, has a training error of 10% and a testing error of 15%. Model B, a complex neural network model, has a training error of 2% but a testing error of 20%. Explain the type of problem Rachel is experiencing with each model, and suggest what she might do to improve the performance of each model.

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